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Check your grasp!

C_4 plants requires 30 ATPs and 12 NADPH + H^+ to synthesize one glucose, but C_3 plants requires only 18 ATPs and 12 NADPH + H^+ to synthesize one glucose molecule. If then, how can you say C_4 plants are more advantageous?

Solution: C_4 plants are more advantageous than C_3 plants because most of the energy lost during photo respiration in C_3 plants.

13.14.3 Significance of C_4 cycle

1. Plants having C_4 cycle are mainly of tropical and sub-tropical regions and are able to survive in environment with low CO_2 concentration.
2. C_4 plants are partially adapted to drought conditions.
3. Oxygen has no inhibitory effect on C_4 cycle since PEP carboxylase is insensitive to O_2.
4. Due to absence of photorespiration, CO_2 Compensation Point for C_4 is lower than that of C_3 plants.

Differences between C_3 Plants (C_3 Cycle) and C_4 Plants (C_4 Cycle) are given in table 13.4.

13.15 Crassulacean Acid Metabolism or CAM cycle

It is one of the carbon pathways identified in succulent plants growing in semi-arid or xerophytic condition. This was first observed in crassulaceae family plants like *Bryophyllum*, *Sedum*, *Kalanchoe* and is the reason behind the name of this cycle. It is also noticed in plants from other families Examples: *Agave*, *Opuntia*, Pineapple and Orchids. The stomata are closed during day and are open during night (Scotoactive). This reverse stomatal rhythm helps to conserve water loss through transpiration and will stop the fixation of CO_2 during the day time. At night time CAM plants fix CO_2

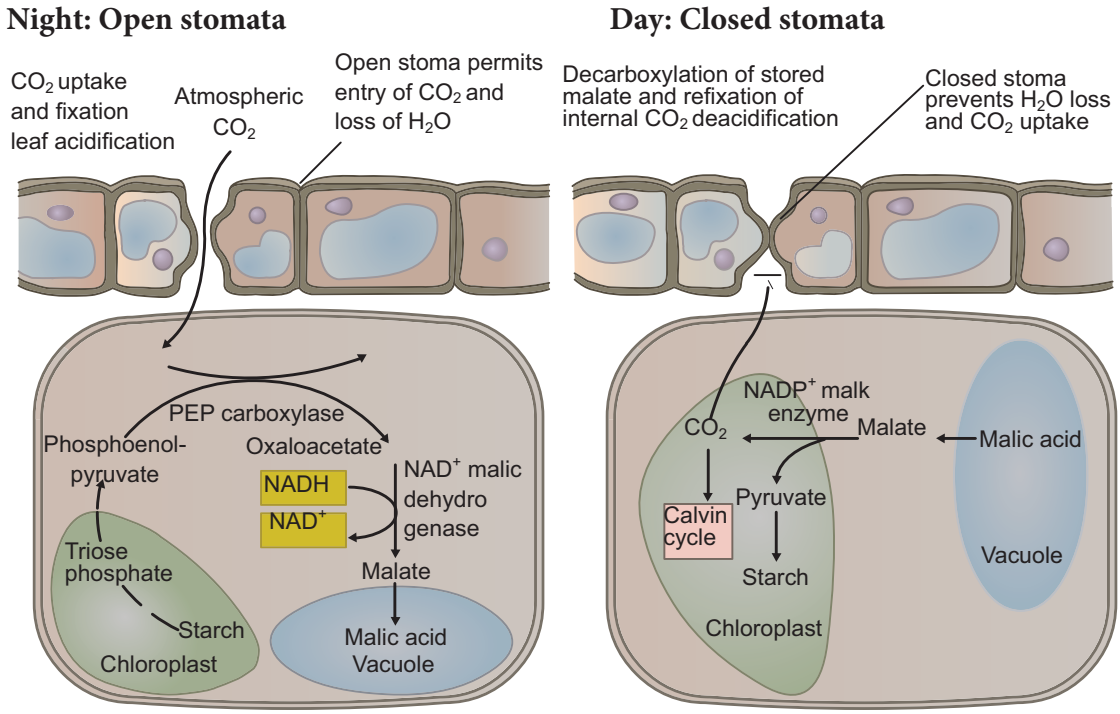


Figure 13.21: CAM cycle